

NARRATIVE REVIEW



Lung- and diaphragm-protective mechanical ventilation in acute respiratory distress syndrome

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Abstract

Lung-protective ventilation is the current standard for mechanical ventilation of patients with acute respiratory distress syndrome (ARDS). Traditionally, this approach has focused on the controlled phase of mechanical ventilation, but emerging data suggest that how patients are managed during assisted ventilation may also impact clinical outcomes. Experimental and observational clinical data indicate that excessive respiratory effort may further damage already injured lungs and may also lead to diaphragm myotrauma. Conversely, insufficient effort and prolonged passive ventilation are associated with diaphragm atrophy and dysfunction. Recent non-invasive techniques to monitor respiratory drive and effort at the bedside have facilitated the development of a new strategy to protect both the lungs and the diaphragm. The lung- and diaphragm-protective (LDP) ventilation framework highlights the need to better integrate ventilation and sedation strategies to facilitate timely and safe spontaneous breathing. This new paradigm has driven the development of emerging supportive and therapeutic modalities, such as diaphragm neurostimulation and partial neuromuscular blockade. Clinical trials are needed to evaluate the impact of LDP strategies on patient-centered outcomes, using designs that account for the possibility of heterogeneity of treatment effect in the ARDS population. In this review, we summarize the physiological background for the LDP framework, as well as the current clinical evidence evaluating this strategy.

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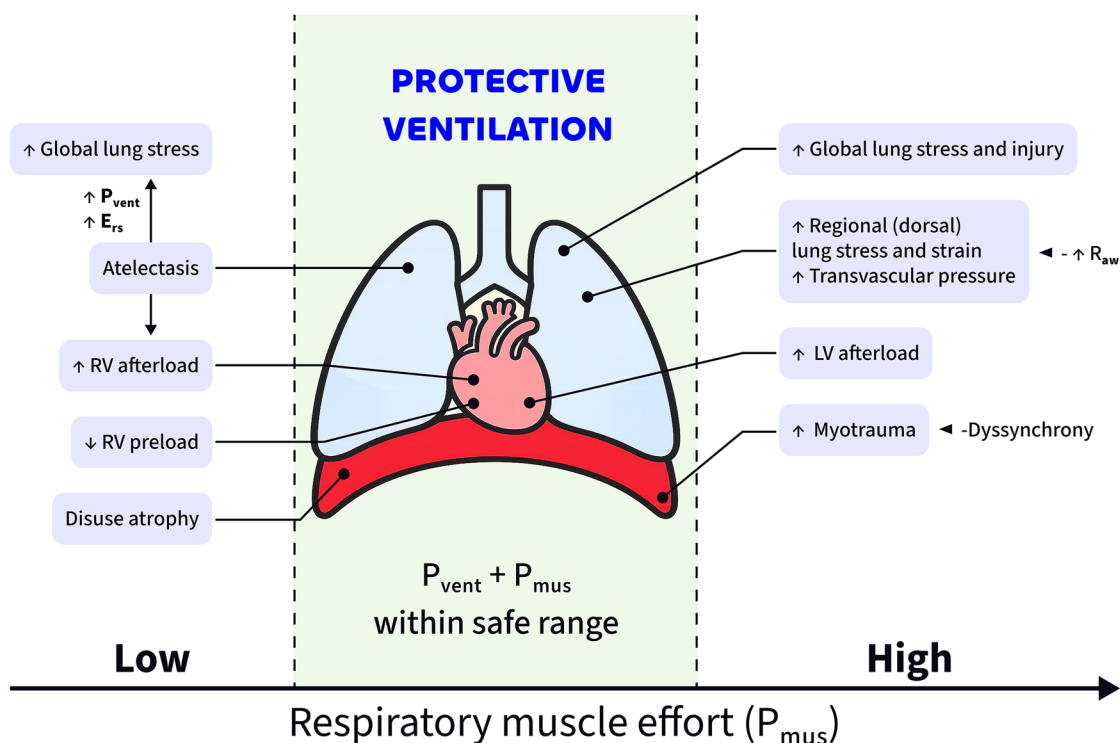
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Taiga Itagaki and Ewan C. Goligher are co-senior.

Visual abstract:

LUNG- AND DIAPHRAGM-PROTECTIVE MECHANICAL VENTILATION IN ACUTE RESPIRATORY DISTRESS SYNDROME

A narrative review



Findings

- Mechanical ventilation can injure both the **lung** and the **diaphragm**.
- **Excessive respiratory effort** promotes lung and diaphragm injury, while **insufficient effort** causes diaphragm atrophy and contributes to lung deflation, atelectasis, and stress.
- The goal is to maintain a "**just right**" level of **spontaneous breathing** when lung injury is more severe: enough to preserve diaphragm function and enhance lung inflation, but not so high as to cause further lung injury.
- Emerging strategies include routine effort monitoring, individualized sedation, partial neuromuscular blockade, ECMO/ECCO₂R, and diaphragm neurostimulation.
- **Future ARDS care may move from lung-protective to lung-and-diaphragm-protective ventilation.**

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Introduction

Mechanical ventilation aims to promote adequate gas exchange, while avoiding ventilator-induced lung injury (VILI). This paradigm gradually developed over the past sixty years based on seminal descriptions of the acute respiratory distress syndrome (ARDS) and VILI [1, 2], followed by demonstration of survival benefit with lung-protective ventilation in clinical trials [3].

Emerging evidence highlights preservation of diaphragmatic function as an important additional target. Extremes of inspiratory effort, whether insufficient or excessive, may be injurious and associated with worse clinical outcomes [4, 5]. These findings have laid the foundation for an emerging approach termed lung- and diaphragm-protective (LDP) ventilation. This strategy builds on the concept of lung-protective ventilation by aiming to integrate three interdependent dimensions: ensuring gas exchange, limiting injurious lung stress, and promoting timely and safe spontaneous breathing. In this review, we synthesize the mechanistic basis of the LDP framework with recent clinical evidence on the feasibility and efficacy of LDP strategies during both controlled and assisted ventilation.

Mechanisms of injury to the lung and diaphragm during mechanical ventilation

Lung injury

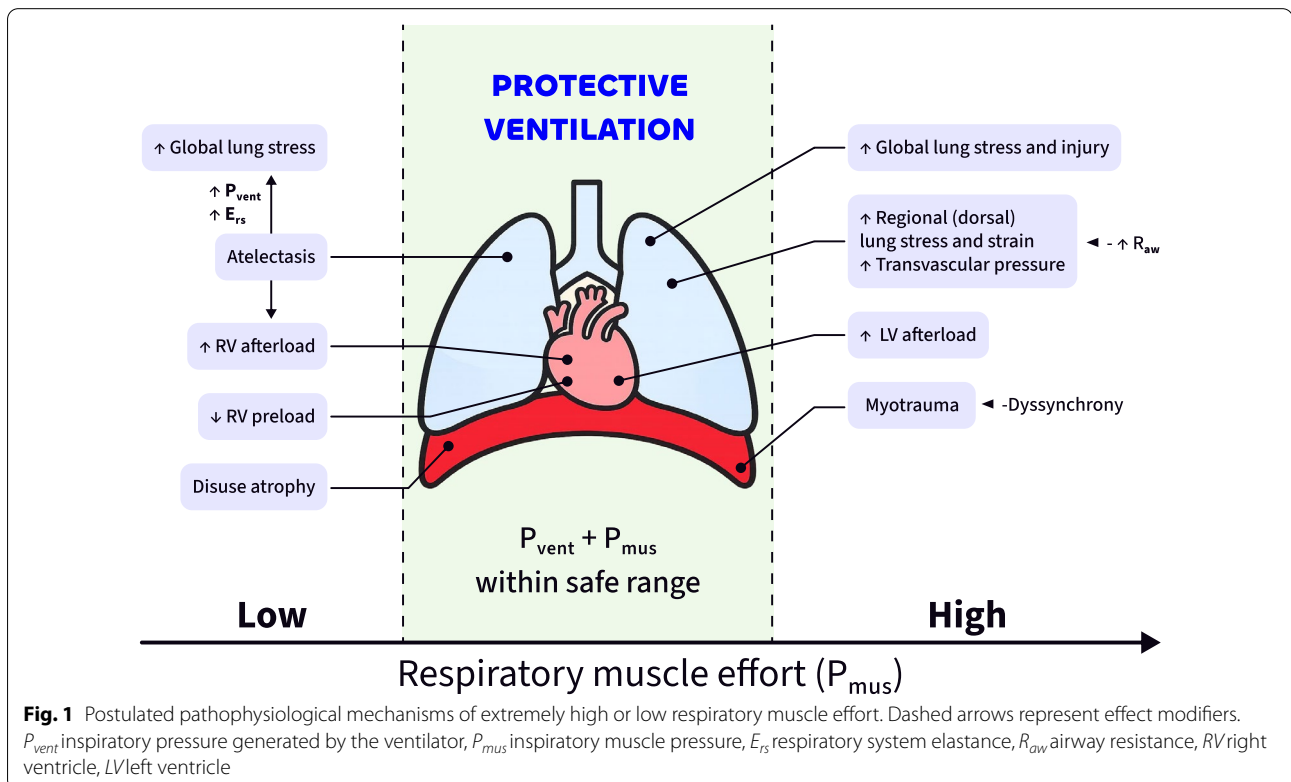
Extensive evidence demonstrates that excessive stress and strain cause lung injury [6, 7]. Several mechanisms

Take-home message

Mechanically ventilated patients with acute respiratory distress syndrome are at an increased risk of both lung and diaphragm injury. Several factors modify this risk, including severity of hypoxemia, respiratory mechanics and dead space. In addition to pressures generated by the ventilator, extremes of respiratory effort may lead to injury and are associated with worse clinical outcomes. Emerging therapeutic strategies may facilitate lung and diaphragm-protective ventilation.

have been proposed, but the final common pathway during controlled ventilation is arguably cyclic overdistension of the “baby lung” (aerated volume at end-expiration) [7, 8].

Limiting cyclic overdistension is challenging during spontaneous breathing, as ARDS patients often exhibit high respiratory drive and effort [9]. Experimental studies indicate that excessive respiratory effort applied to injured lungs can exacerbate damage [10, 11] by generating large pleural pressure swings and increasing global and regional stress (Fig. 1). Lung stress due to excessive effort can increase globally due to excessive tidal volumes, or can be concentrated in dorsal regions adjacent to the diaphragm when the lungs are collapsed and exhibit a solid-like behavior, promoting intrapulmonary air redistribution (pendelluft) and further local overstretch [11, 12]. Intense inspiratory effort promotes tidal



recruitment of dependent lung regions, likely contributing to patient self-inflicted lung injury. It amplifies negative swings in pleural and alveolar pressure, increasing transvascular pressure and left ventricular afterload, and promoting alveolar edema [13, 14]. Increased inspiratory airway resistance may also amplify these mechanical consequences of high inspiratory effort. Elevated central airway resistance increases the inspiratory effort required to generate flow and may facilitate pendelluft, as intrapulmonary gas distribution becomes the path of least resistance [15, 16].

Expiratory muscle activation can exacerbate dorsal lung collapse and further amplify pleural pressure swings through thoracic “spring-loading”, increasing stored elastic recoil prior to inspiration [17] and by enhancing the diaphragm length-tension relationship. Expiratory activity is relatively common in ARDS, being described in up to 50% of patients with moderate-to-severe hypoxemia [17].

Conversely, diaphragm inactivity contributes to atelectasis and loss of lung volume, potentially increasing risk of lung injury [18]. Atelectasis increases right ventricle afterload, and low effort during positive pressure ventilation reduces preload, potentially worsening ventilation-perfusion (V/Q) coupling [19]. Timely restoration of safe and sufficient spontaneous breathing may, therefore, help to protect the lungs, if respiratory effort can be controlled.

Diaphragm injury

Mechanical ventilation can also contribute to diaphragm injury—a phenomenon referred to as myotrauma—resulting in ventilator-induced diaphragm dysfunction (VIDD). Deleterious changes in diaphragm structure and function are common during mechanical ventilation, affecting more than half of mechanically ventilated patients [20, 21]. Myotrauma is mediated by derangements in diaphragm loading conditions during mechanical ventilation [22]. Suppression of respiratory drive due to ventilator overassistance and/or oversedation can lead to excessive inspiratory muscle unloading and rapid diaphragm disuse atrophy and dysfunction (overassistance myotrauma) [23, 24]. Overassistance may occur even with high-level pressure support ventilation, as patient triggering alone does not ensure sufficient diaphragm activation [23, 25]. Conversely, high respiratory drive with associated large inspiratory efforts often reflecting inadequate ventilator assistance can potentially cause load-induced injury (underassistance myotrauma) as shown in experimental studies [26–28], and possibly suggested by clinical studies showing an acute increase in diaphragm thickness and worse outcomes [23, 29]. These factors may

act in combination: a recent experimental study found that diaphragm atrophy induced by overassistance increases susceptibility to load-induced injury if underassistance occurs after transitioning to assisted ventilation [27].

Patient-ventilator dyssynchrony may also contribute to diaphragm dysfunction. Diaphragm lengthening activation (e.g., with reverse triggering or premature cycling) can cause eccentric myotrauma. Risk of injury from eccentric contractions appears to depend on their magnitude and duration [26, 30]. Moderate-to-large eccentric contractions could promote diaphragm dysfunction, whereas low-effort contractions during reverse triggering may preserve diaphragmatic force (in comparison to absent diaphragm activity).

Additionally, diaphragm shortening from application of high PEEP might lead to longitudinal atrophy through sarcomere drop-out, impairing the length-tension relationship of the diaphragm, although clinical relevance of this mechanism remains uncertain [31].

Approach at the bedside

When caring for a patient with ARDS, the clinician's task is to integrate a general knowledge of potential mechanisms of injury and the potential costs of intervention to prevent injury with the unique characteristics of the individual patient. Lung-protective strategies can reduce mortality for patients with high risk for injury, but may often require permissive hypercapnia, increased use of sedation, neuromuscular blockade (NMB), or other interventions with several potential adverse effects. A “one size fits all” approach for ARDS may, therefore, expose patients with low risk of injury to unnecessary interventions, with no net benefit or even potential harm [32, 33]. Factors that determine the risk of lung and diaphragm injury should, therefore, guide implementation of LDP ventilation to optimize outcome for the individual patient.

Assessing the risk of lung and diaphragm injury

The risk of lung injury is largely determined by the size of the lung available for ventilation. Reduced aerated lung volume, reflected by increased respiratory system elastance (E_{rs}), identifies patients in whom sufficient ventilation can only be achieved at the cost of higher distending pressures imposed on the baby lung [34]. The mortality benefit of low tidal volume ventilation varies according to E_{rs} , with higher probability of benefit in patients with high E_{rs} [32]. Similarly, administering NMB may be beneficial if E_{rs} is high, suggesting that controlling respiratory effort may have the greatest benefit in these patients in whom spontaneous breathing effort is more likely to generate injurious lung stress [33].

On the other hand, the requirement for higher tidal volume and respiratory rate is determined largely by dead space and CO_2 production ($\dot{V}\text{CO}_2$). The ventilatory ratio (VR) is a bedside proxy of dead space, and also reflects changes in $\dot{V}\text{CO}_2$ [35]. In combination, E_{rs} and VR may be helpful to predict the probability of maintaining protective targets once respiratory muscle effort is restored, as increases in each could translate into higher respiratory drive and effort [36, 37].

The risk of diaphragm injury also varies between patients. Reduced baseline diaphragm thickness could identify patients with limited contractile reserve and higher risk of diaphragm dysfunction, and has been linked to adverse outcomes [38]. Sepsis, multi-organ failure, and greater baseline severity are associated with accelerated diaphragm catabolism through inflammatory and metabolic pathways, predisposing to rapid muscle atrophy and weakness [39]. The relevance of widely appreciated ARDS biological subphenotypes (hyper- vs. hypoinflammatory) [40] on risk of lung and diaphragm injury remains uncertain. Biological subphenotype did not modify the effect of NMB in a secondary analysis of the ROSE trial [33]. Although higher lung-distending pressures are associated with mortality across both subphenotypes, this association appears stronger in hypoinflammatory ARDS, while excess mortality in hyperinflammatory ARDS may be driven predominantly by extrapulmonary organ dysfunction [41].

Finally, the severity of hypoxemia is correlated with pulmonary inflammation [42], and the risk of superimposed lung injury from ventilation may be higher in patients with more severe hypoxemia. Extremes of respiratory effort and increased lung stress during assisted ventilation, for instance, have only been associated with worse outcome when hypoxemia is moderate-to-severe [4].

These considerations suggest that patients with low E_{rs} , low VR, or mild hypoxemia may be at reduced risk of lung injury, and that it may be safe to liberalize V_T targets and sedation requirements to facilitate spontaneous breathing. We would emphasize that the benefit of management based on this risk framework remains to be demonstrated in clinical trials (see discussion of current evidence below).

Targeting lung protection during controlled ventilation

For patients with moderate or severe lung injury (as reflected by high E_{rs} , high VR, and severe hypoxemia), limiting lung stress should generally take priority over restoring spontaneous diaphragm activity because lung-protective ventilation strategies are associated with established mortality benefit. Interventions should be tailored to lung injury severity as summarized in Fig. 2.

Lowering tidal volume (V_T) reduces lung stress, but applying V_T 6 mL/kg may not be sufficient to ensure lung protection, particularly for patients with markedly reduced lung volumes (high E_{rs}) [6]. Targeting airway driving pressure (ΔP_{aw}), computed as the product between E_{rs} and V_T , allows tailoring V_T to the size of the baby lung, therefore, serving as a more precise surrogate of cyclic lung strain than V_T per predicted body weight [43]. Multiple observational studies confirmed the association between ΔP_{aw} and clinical outcomes, but the potential benefit of targeting ΔP_{aw} as opposed to V_T 6 mL/kg still warrants prospective confirmation in randomized clinical trials. The optimal PEEP strategy remains uncertain, but tailoring PEEP to lung recruitability seems promising [44, 45]. In parallel to applying lung-protective ventilation, interventions to improve E_{rs} and $\dot{V}/Q$ mismatch (e.g., prone positioning), or even to promote gas exchange independent of ventilation through extracorporeal life support (ECLS) should be considered to decrease the risk of lung injury. Future addition of diaphragm neurostimulation may promote enhanced lung protection (through dorsal lung recruitment and decreasing E_{rs}), while also maintaining diaphragm protection by facilitating controlled diaphragmatic activity [46].

Monitoring lung- and diaphragm-protection during assisted ventilation

Once spontaneous breathing resumes, clinicians must determine whether respiratory effort is insufficient, protective, or excessive. Achieving safe levels of respiratory effort may be challenging, as ARDS patients often exhibit high respiratory drive. Monitoring respiratory drive, effort and lung stress during spontaneous breathing is therefore central to the LDP framework. Monitoring should assess both size and synchrony of respiratory efforts. Several techniques to monitor respiratory drive, effort or lung stress have been proposed, including non-invasive techniques using occlusion maneuvers (Table 1).

$P_{0.1}$, the P_{aw} drop during the first 100 ms of an occluded inspiratory effort, is a measure of respiratory drive [47]. It is valid to monitor for inadequate effort [48], but the relationship between respiratory drive and effort varies substantially among patients depending on respiratory muscle strength and inspiratory time. $P_{0.1}$ is therefore only weakly correlated to effort magnitude in both adults and children [49, 50]. If the airway occlusion is maintained for the full duration of the breath (applied at random), the resulting negative swing in airway pressure generated by the patient's effort (P_{occ}) is well correlated with patient effort (both low and high) and can be used to assess dynamic lung stress by estimating the dynamic transpulmonary driving pressure ($\Delta P_{L,dyn,est}$) (Fig. 3) [51]. P_{occ} is validated using the following formulas [50, 51], with P_{occ} obtained as a negative value:

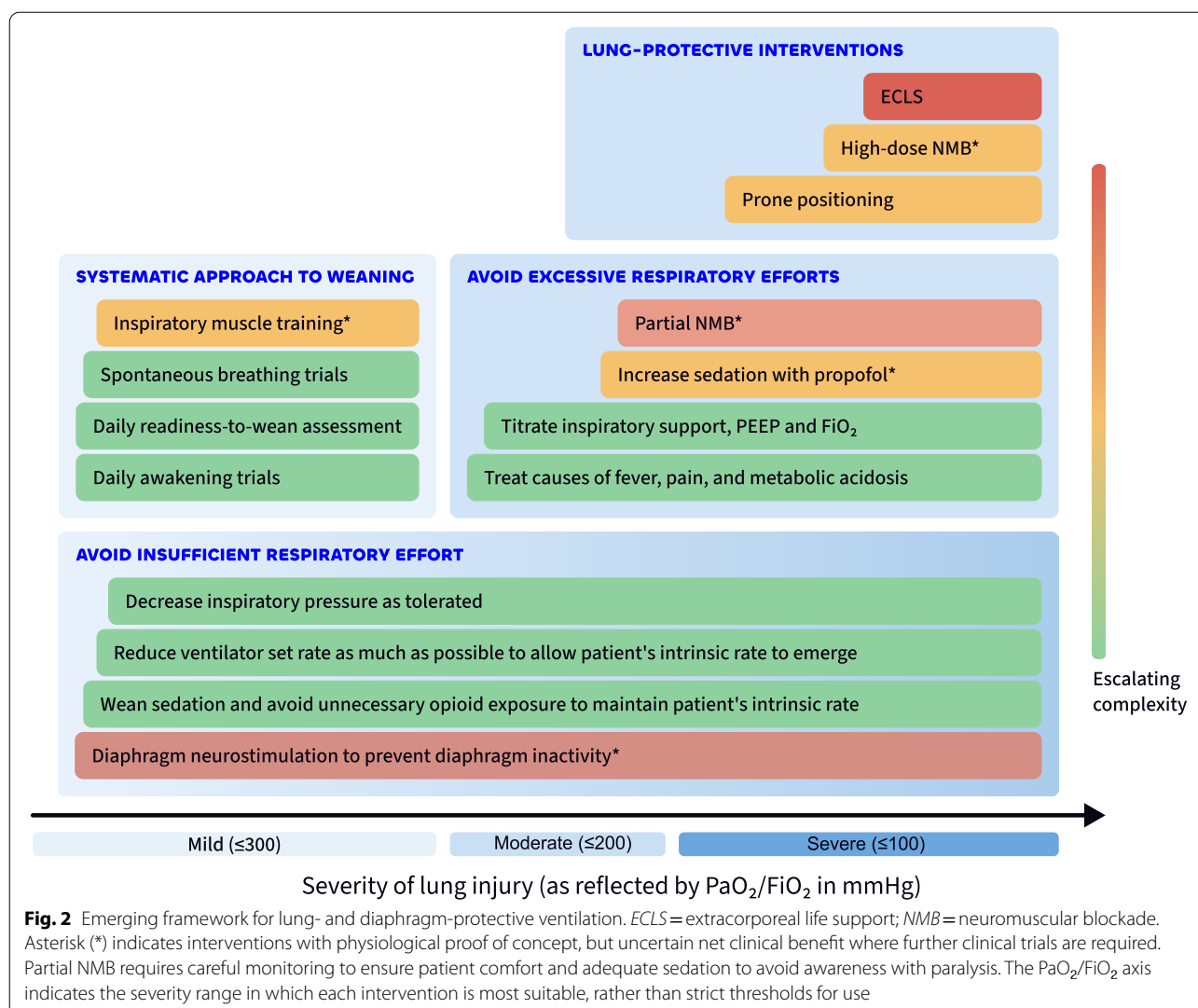
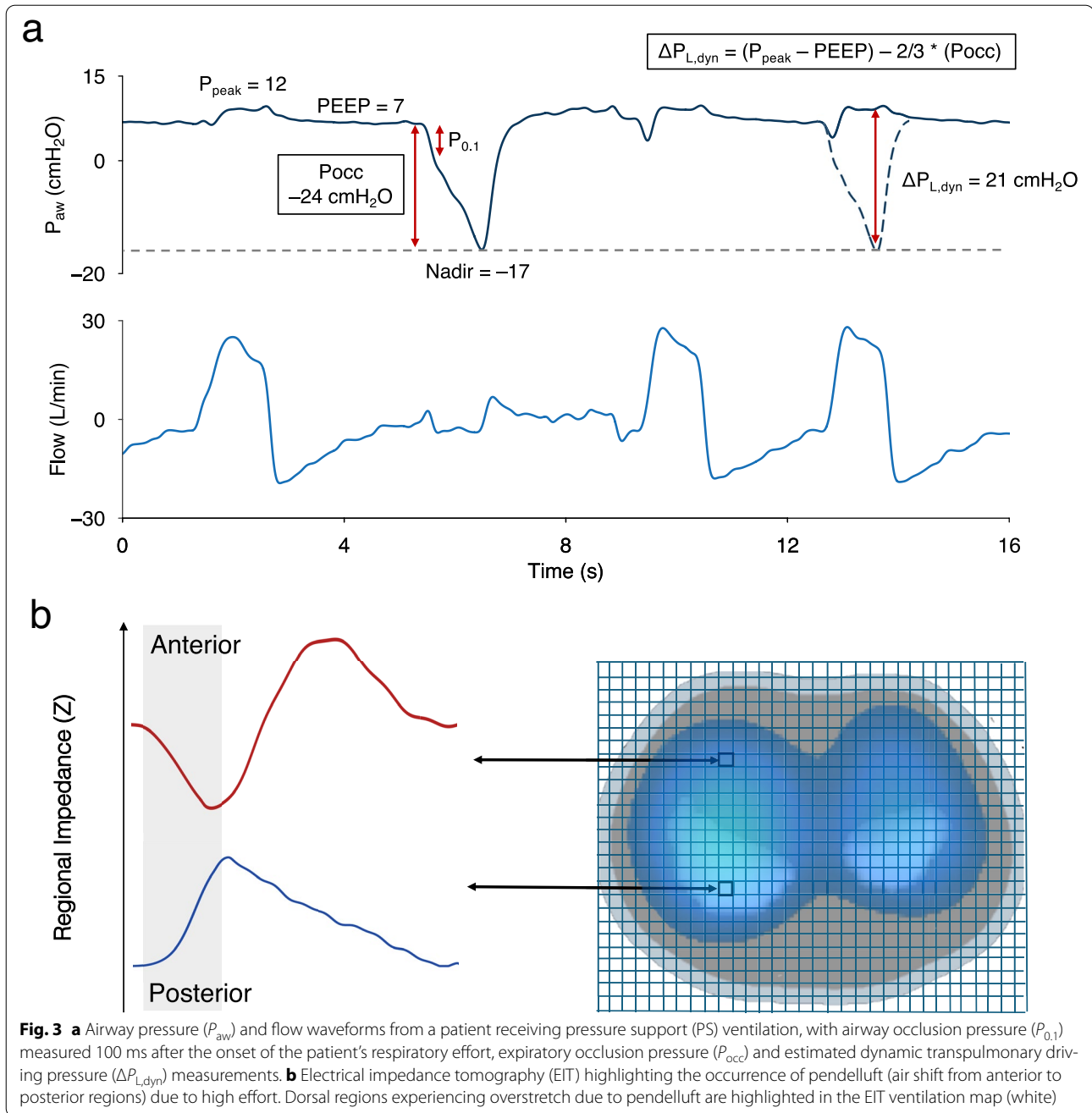


Table 1 Potential targets for lung- and diaphragm-protective strategies

Variable	Lower suggested threshold (to prevent diaphragm atrophy)	Upper suggested threshold (to prevent lung injury or load-induced diaphragm injury)
ΔP_{es}	≥ 3 cmH ₂ O	≤ 8 to 12 cmH ₂ O
$P_{0.1}$	> 1 cmH ₂ O	< 3.5 to 4 cmH ₂ O
P_{occ}	≤ -6 cmH ₂ O	≥ -20 to -15 cmH ₂ O
PMI	> 0 cmH ₂ O	Undefined
$\Delta P_{L,dyn,est}$	Not applicable	≤ 20 to 23 cmH ₂ O
ΔP_{aw}	Not applicable	≤ 12 to 15 cmH ₂ O

ΔP_{es} esophageal pressure swing, $P_{0.1}$ airway occlusion pressure at 100 ms, P_{occ} airway occlusion pressure (whole breath) during an expiratory hold (more negative values indicate higher effort), PMI pressure muscle index, $\Delta P_{L,dyn,est}$ estimated dynamic transpulmonary driving pressure computed as (Peak airway pressure – PEEP) – $2/3 * P_{occ}$ (with P_{occ} as a negative value), ΔP_{aw} airway driving pressure



$$\text{Predicted } P_{mus} = -3/4 \cdot P_{occ} \text{ (adults)}$$

$$\text{Predicted } \Delta P_{es} = 2/3 \cdot P_{occ}$$

$$\text{Predicted } \Delta P_{L,dyn} = (P_{peak} - PEEP) - 2/3 \cdot P_{occ}$$

These equations apply similarly in adults and children. As a dynamic measure reflecting the pressure required

to overcome both elastic and resistive forces in the lung, P_{occ} reflects the risk of dynamic mechanisms of injury, including regional lung stress and low alveolar pressure (dynamic in the sense that they occur within the breath, but are absent during an end-inspiratory hold) [52]. P_{occ} and predicted $\Delta P_{L,dyn}$ are strongly correlated with ICU outcomes [4, 53].

Provided that an inspiratory hold can be reliably performed during assisted ventilation according to quality

criteria [54], plateau pressure can be assessed during spontaneous breathing. The quasi-static airway driving pressure obtained during the hold ($\Delta P_{aw} = P_{plat} - PEEP$) is a measure of global tidal stress on the respiratory system, and is correlated with mortality during assisted ventilation [53]. The pressure muscle index ($PMI = P_{plat} - P_{peak}$) quantifies the pressure generated by the inspiratory muscles to overcome elastic forces to inflation, therefore, being applicable as an estimate of P_{mus} for patients with low airway resistance only [55]. It is moderately correlated with P_{es} -based measures of effort in both adults and children [54, 56] and is useful to detect overassistance (i.e., $PMI < 0$ cmH₂O suggests very low respiratory effort).

If available, esophageal manometry can be used to monitor tidal esophageal pressure swings (ΔP_{es}). When P_{es} is synchronized with airway pressure (P_{aw}), transpulmonary pressure ($P_L = P_{aw} - P_{es}$) can be measured to separate the pressure applied to the lungs from that applied to the chest wall. Dynamic transpulmonary driving pressure ($\Delta P_{L,dyn}$), calculated as the difference between peak inspiratory and end-expiratory P_L , reflects tidal dynamic lung stress. Measured this way, $\Delta P_{L,dyn}$ incorporates the effects of resistance on the pressure inflating the lung. This resistive pressure is not unimportant, and may better reflect risks of high regional lung stress as compared to quasi-static ΔP_L obtained during holds [52].

Targeting lung- and diaphragm-protection during assisted ventilation

If lung protection is achieved within pH targets, interventions to avoid diaphragmatic inactivity while maintaining safe levels of respiratory effort should be initiated. Unnecessary overassistance (passive hyperventilation) should be strictly avoided unless deemed clinically necessary. Reducing the set respiratory rate allows the patient's spontaneous neural rhythm to emerge as sedation is weaned. Reducing sedation and especially avoiding excessive use of opioids (which depress the intrinsic neural respiratory rate) may further help to establish active breathing at diaphragm-protective levels of effort.

Once the patient is breathing spontaneously, respiratory drive, effort, and lung stress should be closely monitored to avoid lung or diaphragm injury. Unless hypoxemia is resolving, excessive respiratory drive and effort should be avoided. If respiratory drive is high, underlying causes, including fever, pain, or metabolic acidosis, should be treated. To implement lung and diaphragm protection during assisted ventilation, clinicians need to be skilled in strategies to modulate and optimize respiratory effort (summarized below) to maintain safe levels of dynamic lung stress and respiratory muscle effort.

As lung injury resolves, a systematic approach to weaning should be implemented, including active sedation weaning, daily readiness-to-wean assessments, and timely spontaneous breathing trials [57]. Diaphragm neurostimulation and inspiratory muscle training could be employed at this point to rehabilitate weakened respiratory muscles, with limited clinical trial evidence to support this [58, 59].

Strategies to modulate effort

Strategies to modulate respiratory effort have been studied extensively in the last few years. These interventions aim to manipulate neural drive, the mechanical load to the lungs, and/or muscle activation, with varying levels of clinical evidence for efficacy.

Modifying ventilator settings

Ventilatory support is often used to reduce the ventilatory load to the respiratory muscles. CO₂ level is one of the main determinants of respiratory drive. Adjusting the ventilator may alter CO₂ clearance and thereby modulate drive. Additionally, ventilatory settings also modulate respiratory drive and effort through non-chemical pathways [60].

Increasing inspiratory pressure support or set tidal volume augments the ventilator's contribution to minute ventilation, and increases lung volume and airflow, potentially reducing respiratory drive and effort. The effect of inspiratory pressure on respiratory effort may depend on the depth of sedation [61]. Titrating PEEP can optimize lung mechanics and V/Q coupling, ultimately reducing the amount of respiratory effort required to achieve adequate gas exchange. PEEP has been shown to reduce effort in an animal model of respiratory failure [11]. However, in the clinical setting, increasing PEEP had variable effects on effort [62]. Higher PEEP reduced both effort and lung-distending pressure in patients in whom increasing PEEP resulted in improved compliance but led to increased effort when resulting in lower compliance by causing overdistention. Therefore, a trial of increasing and decreasing PEEP is warranted to optimize effort. Increasing FiO₂ to achieve higher SpO₂ ($\geq 95\%$) can also reduce respiratory drive, although with variable effect and competing risks of hyperoxemia [63].

Effects of different sedatives

Increasing sedation may be required to reduce excessive effort, although deep sedation entails competing risks, including delirium and long-term disability [64]. Since sedative agents differ in their effects on respiratory drive and breathing pattern, thoughtful agent selection is important to avoid unnecessary escalation of sedation, which may increase adverse effects without effectively

controlling respiratory effort [65]. Propofol, benzodiazepines and inhaled anaesthetics can reduce respiratory drive by different mechanisms, such as blunting the chemoreflex. Propofol has a favorable safety profile, and can reduce effort while preserving respiratory rate [65]. Other agents, such as dexmedetomidine and ketamine, seemingly preserve respiratory drive and effort, being particularly helpful to control agitation while maintaining spontaneous breathing during the weaning phase. Conversely, opioids can decrease respiratory rate, potentially delaying transition to spontaneous breathing by preventing the patient's intrinsic rate from surpassing the ventilator's set rate. Additionally, opioids are also associated with expiratory muscle activation, potentially increasing tidal volume, lung stress, and leading to lung collapse [17]. The opioid-induced breathing pattern (low rate, high tidal volume, abdominal muscle activation), therefore, deviates from that desired for LDP ventilation.

Extracorporeal life support

Increasing sweep gas flow for patients supported with extracorporeal membrane oxygenation (ECMO) may reduce respiratory drive by enhancing CO₂ clearance [66]. In silico modeling suggests extracorporeal CO₂ removal (ECCO₂R) increases the probability of achieving LDP targets in patients with respiratory failure [37]. In the LANDMARK trial, ECMO effectively reduced drive and effort and facilitated achievement of LDP targets compared to mechanical ventilation alone [62]. In a secondary analysis, patients receiving ECMO required less sedation to maintain moderate levels of drive and effort, suggesting CO₂ removal strategies could potentially be used as a sedation-sparing intervention [61].

Partial neuromuscular blockade

In a proof-of-concept study, low doses of rocuronium were administered as a strategy of partial NMB to modulate respiratory muscle activity and decrease transpulmonary pressure and tidal volume, without inducing complete paralysis [67]. In the LANDMARK trial, a subset of patients in whom LDP targets could not be achieved received low dose NMB titrated to reduce effort [62]. All these patients could be maintained within pre-specified targets of LDP. Partial NMB is physiologically attractive because it may reduce tidal lung stress while preserving some diaphragm activation. However, dose requirements are highly individualized, and creating a dissociation between neural drive and achieved effort raises important concerns regarding dyspnea. If partial NMB is used, adequate sedation, careful assessment of comfort, and close monitoring are essential. Its prolonged use should still be regarded as investigational.

Diaphragm neurostimulation

Diaphragm neurostimulation can maintain exogenous diaphragm activity in patients, thereby enhancing pulmonary inflation and cardiac output [68]. This modality can be applied via different routes, e.g., through transcutaneous or transvenous phrenic nerve stimulation. Continuous stimulations can be synchronized with the ventilator's mandatory breaths aiming to preserve diaphragm activity and function, or higher intensity intermittent stimulations (e.g., twice daily) can be applied as a rehabilitation strategy. In a Phase 1 trial, continuous on-demand transvenous diaphragm neurostimulation was shown to be feasible, safe, and efficient to prevent diaphragm inactivity in patients receiving fully controlled mechanical ventilation [69]. Importantly, effort could be kept within the protective range, and no diaphragm atrophy was observed during the 1-week study period, suggesting this strategy could be used to facilitate LDP ventilation in patients who are deeply sedated.

Clinical evidence

The concept of LDP ventilation in ARDS remains a relatively new concept. It is an area of ongoing active investigation, and clinicians should appreciate the strengths and limitations of the emerging evidence base.

Evidence for physiological feasibility and proof of concept

Physiologic and feasibility studies show that respiratory effort and lung-stress targets can often be achieved using combinations of ventilator adjustment, sedation, and extracorporeal support, with partial NMB as a rescue strategy in selected cases (Table 2) [62, 70, 71]. However, most adult studies remain small, short-term, and focused on physiological target attainment rather than patient-centered outcomes. In the single-center physiologic LANDMARK trial, titrating ventilatory settings, sedation and sweep gas flow facilitated achieving of LDP targets in 20 out of 30 patients. Of the remaining 10 patients, 6 received partial NMB and subsequently achieved LDP targets [62]. Furthermore, in a single-center randomized clinical trial, hourly titration of inspiratory pressure support increased the proportion of breaths within protective transdiaphragmatic pressure range [70].

Emerging therapies to enable or sustain LDP ventilation in ARDS are focused on recent and ongoing studies. Early studies of diaphragm neurostimulation demonstrate feasibility in maintaining diaphragm activity and suggest potential benefits for weaning [58, 72]. Partial NMB may facilitate protective ventilation during assisted breathing and, in physiologic studies, improved attainment of protective targets [62, 67]. ECCO₂R can enable ultra-protective ventilation, although its impact on patient-centered outcomes remains uncertain [73, 74].

Table 2 Clinical trials of lung-diaphragm protective strategies and emerging adjuncts

Study	Study Design	Population & Design	Strategy / Intervention	Primary Aim	Key Findings	Key Limitations / Notes
McNamee et al. (2021) (REST) [74]	Multicenter phase III RCT	AHRF (n = 412)	ECCO ₂ R-facilitated lower tidal volumes	Clinical efficacy (mortality)	No mortality benefit	Supports use in research settings
Dres et al. (2025) (RESCUE-3) [72]	Multicenter phase III RCT	Adults failing weaning (n = 216)	Transvenous diaphragm neurostimulation	Clinical efficacy (weaning)	High probability of improved weaning success and shorter ventilation	Early termination; increased serious adverse events
Combes et al. (2019) (SUPERNOVA) [73]	Multicenter phase II single-arm trial	Moderate ARDS; (n = 95)	ECCO ₂ R-facilitated ultra-protective ventilation	Feasibility and safety	Ultra-protective ventilation achieved in most patients	Device-related adverse events; nonrandomized
Khemani et al. (2025) (REDVent) [77]	Single-center phase II RCT	Pediatric ARDS (n = 248)	CDST-guided lung–diaphragm protective ventilation with esophageal manometry	Clinical efficacy (weaning)	Shorter weaning and improved spontaneous breathing trial success in the acute phase	Single center; pediatric population
Costa et al. (2024) (COVEN) [78]	Single-center phase II RCT	Adult COVID-19 ARDS (n = 76)	Extended lung protection with drive control and EIT-guided ventilation	Clinical efficacy (lung recovery)	Faster lung recovery and improved compliance; no mortality difference	Underpowered; pandemic-related early termination
de Vries et al. (2022) (DiaPro) [70]	Single-center phase I RCT	Adult respiratory failure (n = 40)	Inspiratory support titrated to diaphragm effort	Feasibility of diaphragm-protective ventilation	Increased time within diaphragm-protective effort range without worsening lung-protective parameters	Short duration (24 h titration protocol); not powered for clinical outcomes
Mauri et al. (2024) [71]	Multi-center randomized crossover physiologic trial	Adult ARDS on PSV (n = 30)	Personalized PEEP using EIT and transpulmonary pressure	Physiologic efficacy	Reduced dynamic transpulmonary pressure, respiratory drive, and collapse without overdistension	Short-term physiologic outcomes only
Dianti et al. (2022) (LANDMARK) [62]	Single-center, proof-of-concept physiological trial	Adult AHRF; physiologic trial (n = 30)	Effort-targeted ventilation and sedation; partial neuromuscular blockade for refractory effort	Feasibility of achieving lung–diaphragm protective targets	Target attainment improved after titration; partial neuromuscular blockade enhanced probability of achieving protective ranges	Small sample; ECMO-enriched cohort
Morris et al. (2025) (STIMULUS) [58]	Single-center phase I trial	Adult AHRF/postoperative MV (n = 19)	Continuous transvenous diaphragm neurostimulation	Feasibility and safety	Maintained diaphragm activity during passive ventilation; well tolerated	Small sample; short-term feasibility
Doorduyn et al. (2017) [67]	Single-center proof-of-concept physiological trial	Sedated lung injury (n = 10)	Partial neuromuscular blockade during partial support	Physiologic feasibility	Reduced tidal volume and transpulmonary pressure while preserving diaphragm activity	Very small sample; short duration
LANDMARK 2 (ongoing in the PRACTICAL platform) ClinicalTrials.gov: NCT05440851	Multicenter phase II RCT	Adults with AHRF	Lung- and diaphragm-protective ventilation strategy	Feasibility and tolerability	To be determined	To be determined

Table 2 (continued)

Study	Study Design	Population & Design	Strategy / Intervention	Primary Aim	Key Findings	Key Limitations / Notes
PNEUMA (ongoing) ClinicalTrials.gov: NCT04524585	Single-center phase I trial	Adults with moderate-to-severe AHRF	Partial neuromuscular blockade up to 48 h or until resolution of hypoxemia	Feasibility and tolerability	To be determined	To be determined
STAR1 (ongoing) ClinicalTrials.gov: NCT06832306	Multicenter phase II RCT	Adults with AHRF	Transvenous diaphragm neurostimulation to facilitate protective ventilation	Feasibility and tolerability	To be determined	To be determined
STIMT ACTIVATOR 1 (ongoing) ClinicalTrials.gov: NCT05883163	Multicenter phase II RCT	Adults on invasive MV	Transcutaneous diaphragm stimulation to preserve diaphragm trophism	Feasibility and tolerability	To be determined	To be determined
IMPROV (ongoing in the PRACTICAL platform) ClinicalTrials.gov: NCT05440851	Multicenter pilot RCT	Adults with AHRF	Inspiratory muscle training to accelerate recovery from respiratory failure	Feasibility and tolerability	To be determined	To be determined

Novel CO₂ clearance and drive-modulating approaches, such as “Through-Flow” ventilation applying higher tracheal gas flow during inspiration, may further reduce injurious effort and lung stress [75].

Observational outcome studies

Multiple cohort studies demonstrated that high and low levels of respiratory drive and effort are associated with increased mortality, especially in the presence of more severe hypoxemia [4, 53, 76]. Higher lung-distending pressures during spontaneous breathing are also associated with increased mortality. These consistent associations across independent cohorts, together with mechanistic evidence of injury at both extremes of effort, underscore the potential clinical relevance of targeting respiratory effort during assisted ventilation.

Outcomes in clinical trials

In the pediatric REDvent randomized trial, LDP ventilation guided by a decision support tool focused on systematic adjustment of inspiratory pressure to keep esophageal pressure in a target range shortened ventilator weaning time [77], while the COVEN phase II trial observed faster lung recovery with extended lung-protective strategies in adults with COVID-19 ARDS [78]. Together, these studies support feasibility and highlight the need for definitive outcome-focused trials.

Future directions

The validation of readily accessible non-invasive tools that can quantify respiratory effort and lung-distending pressure has supported embedding routine monitoring into clinical practice [79]. Continuous non-invasive monitoring of respiratory effort and patient-ventilator synchrony is emerging through physiology- or artificial intelligence-based estimation [80, 81]. Further advances in monitoring techniques such as electrical impedance tomography (EIT), to assess regional lung stress might facilitate evaluating lung protection during assisted ventilation [82]. Evidence-based recommendations linking monitoring to interventions that improve patient outcomes remain essential. Research should now focus on validating therapeutic targets and evaluating implementable strategies that account for patient heterogeneity.

In this regard, clinical decision support tools may facilitate early identification and risk stratification of patients susceptible to lung and diaphragm injury [83]. More advanced systems, such as digital twins and artificial intelligence, can integrate physiological data to inform which strategies are most likely to succeed for a given patient [84].

The armamentarium to deliver LDP ventilation is also likely to expand. Techniques such as partial NMB, ECCO₂R and diaphragm neurostimulation have shown promise in early work [58, 62, 67, 85]. Further studies are required to define appropriate patient subgroups, establish safe implementation, optimal treatment protocols, and demonstrate meaningful clinical benefit. Nonetheless, future studies should account for the broad spectrum of pathophysiological phenotypes and anticipate heterogeneity of treatment effect—understanding these patient-level effect modifiers will enable clinicians to individualize prioritization when LDP goals conflict. Adaptive trial designs using predictive and prognostic enrichment should be considered regardless of the intervention being tested [86].

In parallel, the impact of mechanical ventilation on other organs is still being understood. Ventilator-associated brain injury (VABI) is an emerging concept describing the direct negative impact of passive ventilation on the brain, independent of sedation and illness severity, pointing toward a new frontier for investigation in ARDS [87].

Conclusion

Lung- and diaphragm-protective ventilation represents an evolution of traditional lung-protective strategies in ARDS, integrating control of lung stress with preservation of appropriate respiratory effort. Clinical data consistently demonstrate that excessive and insufficient effort, as well as elevated lung-distending pressures during assisted ventilation, are associated with worse outcomes. Emerging monitoring tools and early-phase interventional studies suggest that targeting respiratory effort and lung stress is feasible. Definitive trials are now required to determine whether systematically implementing LDP strategies can improve survival and long-term recovery in ARDS.

Abbreviations

ECCO₂-R extracorporeal carbon dioxide removal, *RCT* randomized clinical trial, *AHRF* acute hypoxemic respiratory failure, *ECMO* extracorporeal membrane oxygenation, *EIT* electrical impedance tomography, *PSV* pressure support ventilation, *CDS* clinical decision support tool, *MV* mechanical ventilation

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Data availability

Not applicable.

Declarations

Conflicts of interest

ECG is supported by an Early Career Investigator Award from the National Sanitarium Association and received personal fees for consulting or speaking from Baxter, Vyair, Dräger, Getinge, Lungpacer, Stimit, and Fisher & Paykel. DLG has received payments for travel expenses by Getinge, Draeger, personal fees by Draeger, GE, Nihon Kohden and Fisher and Paykel, and research grants by Fisher and Paykel and GE. MBPA received grants from Timpel, Magnamed, Medtronic, and Nihon Kohden. AJ received research funding (paid to the lab), from ZonMw, Liberate Medical, Pulmotech BV, Health ~ Holland and Finapres. LB’s laboratory has received grants from Medtronic, Vitalaire and Stimit, equipment from Fisher Paykel, Stimit, Sentec and Cerebra. LB has received personal fees from Stimit (trial) and Fisher Paykel (symposium). The other authors have no conflicts of interest to declare.

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References

- Ashbaugh DG, Bigelow DB, Petty TL, Levine BE (1967) Acute respiratory distress in adults. *Lancet* 2:319–323. [https://doi.org/10.1016/s0140-6736\(67\)90168-7](https://doi.org/10.1016/s0140-6736(67)90168-7)
- Webb HH, Tierney DF (1974) Experimental pulmonary edema due to intermittent positive pressure ventilation with high inflation pressures. Protection by positive end-expiratory pressure. *Am Rev Respir Dis* 110:556–565. <https://doi.org/10.1164/arrd.1974.110.5.556>
- The Acute Respiratory Distress Syndrome Network (2000) Ventilation with lower tidal volumes as compared with traditional tidal volumes for acute lung injury and the acute respiratory distress syndrome. *N Engl J Med* 342:1301–1308. <https://doi.org/10.1056/NEJM200005043421801>
- Dianti J, Lovblom LE, Iftikhar MA et al (2025) Association of respiratory drive and effort with mortality and time to discharge in patients on mechanical ventilation in Canada: a longitudinal, prospective, registry-based cohort study. *Lancet Respir Med* 13:1087–1095. [https://doi.org/10.1016/S2213-2600\(25\)00297-8](https://doi.org/10.1016/S2213-2600(25)00297-8)
- Parrilla-Gómez FJ, Castellvi-Font A, Boutonnet V et al (2025) Association of breathing effort with survival in patients with acute respiratory distress syndrome. *Crit Care Med* 53:e1982–e1994. <https://doi.org/10.1097/CCM.00000000000006797>
- Terragni PP, Rosboch G, Tealdi A et al (2007) Tidal hyperinflation during low tidal volume ventilation in acute respiratory distress syndrome. *Am J Respir Crit Care Med* 175:160–166. <https://doi.org/10.1164/rccm.200607-9150C>
- Protti A, Andreis DT, Monti M et al (2013) Lung stress and strain during mechanical ventilation: any difference between statics and dynamics? *Crit Care Med* 41:1046–1055. <https://doi.org/10.1097/CCM.0b013e31827417a6>
- Plens GM, Costa ELV, Amato MBP (2026) Ventilator-induced lung injury: between tidal recruitment, overdistension and the baby lung paradigm. *Intensive Care Med*. <https://doi.org/10.1007/s00134-026-08313-9>
- Vaporidi K, Akoumianaki E, Telias I et al (2020) Respiratory drive in critically ill patients. pathophysiology and clinical implications. *Am J Respir Crit Care Med* 201:20–32. <https://doi.org/10.1164/rccm.201903-0596SO>
- Yoshida T, Uchiyama A, Matsuura N et al (2013) The comparison of spontaneous breathing and muscle paralysis in two different severities of experimental lung injury*. *Crit Care Med* 41:536–545. <https://doi.org/10.1097/CCM.0b013e3182711972>
- Morais CCA, Koyama Y, Yoshida T et al (2018) High positive end-expiratory pressure renders spontaneous effort noninjurious. *Am J Respir Crit Care Med* 197:1285–1296. <https://doi.org/10.1164/rccm.201706-1244OC>
- Yoshida T, Torsani V, Gomes S et al (2013) Spontaneous effort causes occult pendelluft during mechanical ventilation. *Am J Respir Crit Care Med* 188:1420–1427. <https://doi.org/10.1164/rccm.201303-0539OC>
- Telias I, Madorno M, Pham T et al (2025) Physiological consequences of breathing effort according to the mode of ventilation during acute hypoxemic respiratory failure. *Am J Respir Crit Care Med*. <https://doi.org/10.1164/rccm.202411-2155OC>
- Kallet RH, Alonso JA, Luce JM, Matthay MA (1999) Exacerbation of acute pulmonary edema during assisted mechanical ventilation using a low-tidal volume, lung-protective ventilator strategy. *Chest* 116:1826–1832. <https://doi.org/10.1378/chest.116.6.1826>
- Castellvi-Font A, Goligher EC (2026) Reconsidering resistance: a potentially overlooked risk factor for lung injury during assisted ventilation. *Am J Respir Crit Care Med*. <https://doi.org/10.1093/ajrccm/aamag067>
- Grieco DL, Delle Cese L, Menga LS et al (2023) Physiological effects of awake prone position in acute hypoxemic respiratory failure. *Crit Care* 27:315. <https://doi.org/10.1186/s13054-023-04600-9>
- Plens GM, Droghi MT, Alcalá GC et al (2024) Expiratory muscle activity counteracts positive end-expiratory pressure and is associated with fentanyl dose in patients with acute respiratory distress syndrome. *Am J Respir Crit Care Med* 209:563–572. <https://doi.org/10.1164/rccm.202308-1376OC>
- Rohrs EC, Bassi TG, Fernandez KC et al (2021) Diaphragm neurostimulation during mechanical ventilation reduces atelectasis and transpulmonary plateau pressure, preserving lung homogeneity and PaO₂/FIO₂. *J Appl Physiol* (1985) 131:290–301. <https://doi.org/10.1152/jappphysiol.00119.2021>
- Slobod D, Yusuff H, Zochios V, Protecting the Right Ventricle network (PRORVnet) (2026) Right ventricular-protective ventilation in ARDS: thinking beyond lung protection. *Intensive Care Med*. <https://doi.org/10.1007/s00134-026-08421-6>
- Goligher EC, Dres M, Fan E et al (2018) Mechanical ventilation–induced diaphragm atrophy strongly impacts clinical outcomes. *Am J Respir Crit Care Med* 197:204–213. <https://doi.org/10.1164/rccm.201703-0536OC>
- Dres M, Dubé B-P, Mayaux J et al (2017) Coexistence and impact of limb muscle and diaphragm weakness at time of liberation from mechanical ventilation in medical intensive care unit patients. *Am J Respir Crit Care Med* 195:57–66. <https://doi.org/10.1164/rccm.201602-0367OC>
- Powers SK, Wiggs MP, Sollanek KJ, Smuder AJ (2013) Ventilator-induced diaphragm dysfunction: cause and effect. *Am J Physiol Regul Integr Comp Physiol* 305:R464–477. <https://doi.org/10.1152/ajpregu.00231.2013>
- Goligher EC, Fan E, Herridge MS et al (2015) Evolution of diaphragm thickness during mechanical ventilation. impact of inspiratory effort. *Am J Respir Crit Care Med* 192:1080–1088. <https://doi.org/10.1164/rccm.201503-0620OC>
- Jaber S, Petrof BJ, Jung B et al (2011) Rapidly progressive diaphragmatic weakness and injury during mechanical ventilation in humans. *Am J Respir Crit Care Med* 183:364–371. <https://doi.org/10.1164/rccm.201004-0670OC>
- Hudson MB, Smuder AJ, Nelson WB et al (2012) Both high level pressure support ventilation and controlled mechanical ventilation induce diaphragm dysfunction and atrophy. *Crit Care Med* 40:1254–1260. <https://doi.org/10.1097/CCM.0b013e31823c8cc9>
- Damiani LF, Engelberts D, Bastia L et al (2022) Impact of reverse triggering dyssynchrony during lung-protective ventilation on diaphragm function: an experimental model. *Am J Respir Crit Care Med* 205:663–673. <https://doi.org/10.1164/rccm.202105-1089OC>
- Capdevila M, Pensier J, De Jong A et al (2025) Impact of underassisted ventilation on diaphragm function and structure in a porcine model. *Anesthesiology* 142:896–906. <https://doi.org/10.1097/ALN.0000000000005390>
- Ebihara S, Hussain SNA, Danialou G et al (2002) Mechanical ventilation protects against diaphragm injury in sepsis: interaction of oxidative and mechanical stresses. *Am J Respir Crit Care Med* 165:221–228. <https://doi.org/10.1164/ajrccm.165.2.2108041>
- Goligher EC, Brochard LJ, Reid WD et al (2019) Diaphragmatic myotrauma: a mediator of prolonged ventilation and poor patient outcomes in acute respiratory failure. *Lancet Respir Med* 7:90–98. [https://doi.org/10.1016/S2213-2600\(18\)30366-7](https://doi.org/10.1016/S2213-2600(18)30366-7)
- Coiffard B, Dianti J, Telias I et al (2024) Dyssynchronous diaphragm contractions impair diaphragm function in mechanically ventilated patients. *Crit Care* 28:107. <https://doi.org/10.1186/s13054-024-04894-3>
- Lindqvist J, van den Berg M, van der Pijl R et al (2018) Positive end-expiratory pressure ventilation induces longitudinal atrophy in diaphragm fibers. *Am J Respir Crit Care Med* 198:472–485. <https://doi.org/10.1164/rccm.201709-1917OC>
- Goligher EC, Costa ELV, Yarnell CJ et al (2021) Effect of lowering V_t on mortality in acute respiratory distress syndrome varies with respiratory system elastance. *Am J Respir Crit Care Med* 203:1378–1385. <https://doi.org/10.1164/rccm.202009-3536OC>
- Zalucky AA, Dianti J, Neyton LPA et al (2025) Elastance may determine the neuromuscular blockade effect on mortality in acute respiratory distress syndrome. *Am J Respir Crit Care Med* 211:966–974. <https://doi.org/10.1164/rccm.202406-1231OC>
- Catozzi G, Pozzi T, Nocera D et al (2025) Rethinking ARDS classification: oxygenation impairment fails to predict VILI risk. *Intensive Care Med* 51:62–71. <https://doi.org/10.1007/s00134-024-07712-0>
- Sinha P, Calfee CS, Beitler JR et al (2019) Physiologic analysis and clinical performance of the ventilatory ratio in acute respiratory distress syndrome. *Am J Respir Crit Care Med* 199:333–341. <https://doi.org/10.1164/rccm.201804-0692OC>
- Vaporidi K, Psarologakis C, Prokhou A et al (2019) Driving pressure during proportional assist ventilation: an observational study. *Ann Intensive Care* 9:1. <https://doi.org/10.1186/s13613-018-0477-4>
- Ratano D, Zhang B, Dianti J et al (2024) Lung- and diaphragm-protective strategies in acute respiratory failure: an in silico trial. *Intensive Care Med Exp* 12:20. <https://doi.org/10.1186/s40635-024-00606-x>
- Sklar MC, Dres M, Fan E et al (2020) Association of low baseline diaphragm muscle mass with prolonged mechanical ventilation and

- mortality among critically ill adults. *JAMA Netw Open* 3:e1921520. <https://doi.org/10.1001/jamanetworkopen.2019.21520>
39. Rocheteau P, Chatre L, Briand D et al (2015) Sepsis induces long-term metabolic and mitochondrial muscle stem cell dysfunction amenable by mesenchymal stem cell therapy. *Nat Commun* 6:10145. <https://doi.org/10.1038/ncomms10145>
 40. Matthay MA, Arabi YM, Siegel ER et al (2020) Phenotypes and personalized medicine in the acute respiratory distress syndrome. *Intensive Care Med* 46:2136–2152. <https://doi.org/10.1007/s00134-020-06296-9>
 41. Pensier J, Fosset M, Paschold B-S et al (2026) Effects of inflammatory phenotypes in acute respiratory distress syndrome on mortality and partitioning of lung and chest wall mechanics in patients in the USA and Canada: a retrospective cohort study. *Lancet Respir Med*. [https://doi.org/10.1016/S2213-2600\(26\)00045-7](https://doi.org/10.1016/S2213-2600(26)00045-7)
 42. Calfee CS, Ware LB, Eisner MD et al (2008) Plasma receptor for advanced glycation end products and clinical outcomes in acute lung injury. *Thorax* 63:1083–1089. <https://doi.org/10.1136/thx.2008.095588>
 43. Amato MBP, Meade MO, Slutsky AS et al (2015) Driving pressure and survival in the acute respiratory distress syndrome. *N Engl J Med* 372:747–755. <https://doi.org/10.1056/NEJMsa1410639>
 44. Chen L, Del Sorbo L, Grieco DL et al (2020) Potential for lung recruitment estimated by the recruitment-to-inflation ratio in acute respiratory distress syndrome: a clinical trial. *Am J Respir Crit Care Med* 201:178–187. <https://doi.org/10.1164/rccm.201902-0334OC>
 45. Yuan X, Zhong M, Li Z et al (2026) Electrical impedance tomography-guided positive end-expiratory pressure and mortality of patients with the acute respiratory distress syndrome: the EITVent randomized clinical trial. *Am J Respir Crit Care Med* 212:440–451. <https://doi.org/10.1093/ajrccm/aamaf125>
 46. Dres M, Heunks L, Goligher EC (2026) The emerging role of diaphragm neurostimulation in ARDS. *Intensive Care Med*. <https://doi.org/10.1007/s00134-026-08443-0>
 47. Whitelaw WA, Derenne JP, Milic-Emili J (1975) Occlusion pressure as a measure of respiratory center output in conscious man. *Respir Physiol* 23:181–199. [https://doi.org/10.1016/0034-5687\(75\)90059-6](https://doi.org/10.1016/0034-5687(75)90059-6)
 48. Telias I, Junhasavasdikul D, Rittayamai N et al (2020) Airway occlusion pressure as an estimate of respiratory drive and inspiratory effort during assisted ventilation. *Am J Respir Crit Care Med* 201:1086–1098. <https://doi.org/10.1164/rccm.201907-1425OC>
 49. Smits FE, Rietveld PJ, Snoep JWM et al (2025) P0.1 is an unreliable measure of effort in support mechanical ventilation in comparison with esophageal-derived measures of effort: a comparison study. *Crit Care Med* 53:e1650–e1658. <https://doi.org/10.1097/CCM.00000000000006745>
 50. Ito Y, Herrera MG, Hotz JC et al (2023) Estimation of inspiratory effort using airway occlusion maneuvers in ventilated children: a secondary analysis of an ongoing randomized trial testing a lung and diaphragm protective ventilation strategy. *Crit Care* 27:466. <https://doi.org/10.1186/s13054-023-04754-6>
 51. Bertoni M, Telias I, Urner M et al (2019) A novel non-invasive method to detect excessively high respiratory effort and dynamic transpulmonary driving pressure during mechanical ventilation. *Crit Care* 23:346. <https://doi.org/10.1186/s13054-019-2617-0>
 52. Zhang X, Bassi T, Dianti J et al (2025) Monitoring dynamic mechanisms of lung injury risk during spontaneous breathing in mechanically ventilated patients. *Am J Respir Crit Care Med*. <https://doi.org/10.1164/rccm.202504-0838RL>
 53. Grassi A, Bianchi I, Teggia Droghi M et al (2025) Increased driving pressure during assisted ventilation for hypoxemic respiratory failure is associated with lower ICU survival: the ICEBERG study. *Am J Respir Crit Care Med*. <https://doi.org/10.1164/rccm.202411-2146OC>
 54. Bianchi I, Grassi A, Pham T et al (2022) Reliability of plateau pressure during patient-triggered assisted ventilation. Analysis of a multicentre database. *J Crit Care* 68:96–103. <https://doi.org/10.1016/j.jccr.2021.12.002>
 55. Foti G, Cereda M, Banfi G et al (1997) End-inspiratory airway occlusion: a method to assess the pressure developed by inspiratory muscles in patients with acute lung injury undergoing pressure support. *Am J Respir Crit Care Med* 156:1210–1216. <https://doi.org/10.1164/ajrccm.156.4.96-02031>
 56. Kyogoku M, Shimatani T, Hotz JC et al (2021) Direction and magnitude of change in plateau from peak pressure during inspiratory holds can identify the degree of spontaneous effort and elastic workload in ventilated patients. *Crit Care Med* 49:517–526. <https://doi.org/10.1097/CCM.00000000000004746>
 57. Blackwood B, Burns KE, Cardwell CR, O'Halloran P (2014) Protocolized versus non-protocolized weaning for reducing the duration of mechanical ventilation in critically ill adult patients. *Cochrane Database Syst Rev* 6:CD006904. <https://doi.org/10.1002/14651858.CD006904.pub3>
 58. Morris IS, Bassi T, Bellissimo CA et al (2025) Continuous on-demand diaphragm neurostimulation to prevent diaphragm inactivity during mechanical ventilation: a phase 1 clinical trial (STIMULUS). *Am J Respir Crit Care Med* 211:1442–1451. <https://doi.org/10.1164/rccm.202407-1483OC>
 59. Vorona S, Sabatini U, Al-Maqbali S et al (2018) Inspiratory muscle rehabilitation in critically ill adults: a systematic review and meta-analysis. *Ann Am Thorac Soc* 15:735–744. <https://doi.org/10.1513/AnnalsATS.201712-961OC>
 60. Georgopoulos D, Roussos C (1996) Control of breathing in mechanically ventilated patients. *Eur Respir J* 9:2151–2160. <https://doi.org/10.1183/09031936.96.09102151>
 61. Dianti J, Morris IS, Bassi TG et al (2024) Sedation-ventilation interaction in acute hypoxemic respiratory failure. *Chest Crit Care* 2:100067. <https://doi.org/10.1016/j.chstcc.2024.100067>
 62. Dianti J, Fard S, Wong J et al (2022) Strategies for lung- and diaphragm-protective ventilation in acute hypoxemic respiratory failure: a physiological trial. *Crit Care* 26:259. <https://doi.org/10.1186/s13054-022-04123-9>
 63. Pesenti A, Rossi N, Calori A et al (1993) Effects of short-term oxygenation changes on acute lung injury patients undergoing pressure support ventilation. *Chest* 103:1185–1189. <https://doi.org/10.1378/chest.103.4.1185>
 64. Wongtangman K, Borngaesser F, Rudolph MI et al (2026) Association of medication-induced deep sedation and emotional distress during mechanical ventilation with loss of independent living: an observational cohort study. *Lancet Respir Med* 14:49–59. [https://doi.org/10.1016/S2213-2600\(25\)00264-4](https://doi.org/10.1016/S2213-2600(25)00264-4)
 65. Quickfall D, Sklar MC, Tomlinson G et al (2024) The influence of drugs used for sedation during mechanical ventilation on respiratory pattern during unassisted breathing and assisted mechanical ventilation: a physiological systematic review and meta-analysis. *EClinicalMedicine* 68:102417. <https://doi.org/10.1016/j.eclinm.2023.102417>
 66. Crotti S, Bottino N, Ruggeri GM et al (2017) Spontaneous breathing during extracorporeal membrane oxygenation in acute respiratory failure. *Anesthesiology* 126:678–687. <https://doi.org/10.1097/ALN.0000000000001546>
 67. Doorduyn J, Nolle JL, Roesthuis LH et al (2017) Partial neuromuscular blockade during partial ventilatory support in sedated patients with high tidal volumes. *Am J Respir Crit Care Med* 195:1033–1042. <https://doi.org/10.1164/rccm.201605-1016OC>
 68. Morris IS, Bassi T, Castellvi-Font A et al (2026) Diaphragm neurostimulation mitigates the adverse cardiopulmonary effects of positive pressure ventilation. *Crit Care*. <https://doi.org/10.1186/s13054-026-05870-9>
 69. Morris IS, Bassi T, Bellissimo CA et al (2023) Proof of concept for continuous on-demand phrenic nerve stimulation to prevent diaphragm disuse during mechanical ventilation (STIMULUS): a phase 1 clinical trial. *Am J Respir Crit Care Med* 208:992–995. <https://doi.org/10.1164/rccm.202305-0791LE>
 70. De Vries HJ, Jonkman AH, De Grooth HJ et al (2022) Lung- and diaphragm-protective ventilation by titrating inspiratory support to diaphragm effort: a randomized clinical trial. *Crit Care Med* 50:192–203. <https://doi.org/10.1097/CCM.00000000000005395>
 71. Mauri T, Grieco DL, Spinelli E et al (2024) Personalized positive end-expiratory pressure in spontaneously breathing patients with acute respiratory distress syndrome by simultaneous electrical impedance tomography and transpulmonary pressure monitoring: a randomized crossover trial. *Intensive Care Med* 50:2125–2137. <https://doi.org/10.1007/s00134-024-07695-y>
 72. Dres M, Ewert R, Conrad SA et al (2025) Temporary transvenous diaphragm neurostimulation for weaning from mechanical ventilation

- (RESCUE-3). *Am J Respir Crit Care*. <https://doi.org/10.1164/rccm.202505-1056OC>
73. Combes A, Fanelli V, Pham T et al (2019) Feasibility and safety of extracorporeal CO₂ removal to enhance protective ventilation in acute respiratory distress syndrome: the SUPERNOVA study. *Intensive Care Med* 45:592–600. <https://doi.org/10.1007/s00134-019-05567-4>
 74. McNamee JJ, Gillies MA, Barrett NA et al (2021) Effect of lower tidal volume ventilation facilitated by extracorporeal carbon dioxide removal vs standard care ventilation on 90-day mortality in patients with acute hypoxemic respiratory failure: the REST randomized clinical trial. *JAMA* 326:1013. <https://doi.org/10.1001/jama.2021.13374>
 75. Sinderby C, Bassi T, Liu L et al (2026) Integration of CO₂ clearance and continuous neurally adjusted ventilatory assist in an animal model of respiratory distress. *Respir Physiol Neurobiol* 339:104499. <https://doi.org/10.1016/j.resp.2025.104499>
 76. Bellani G, Grassi A, Sosio S et al (2019) Driving pressure is associated with outcome during assisted ventilation in acute respiratory distress syndrome. *Anesthesiology* 131:594–604. <https://doi.org/10.1097/ALN.0000000000002846>
 77. Khemani RG, Bhalla A, Hotz JC et al (2025) randomized trial of lung and diaphragm protective ventilation in children. *NEJM Evid 4:EVI-Doa2400360*. <https://doi.org/10.1056/EVIDoA2400360>
 78. Costa ELV, Alcalá GC, Tucci MR et al (2024) Impact of extended lung protection during mechanical ventilation on lung recovery in patients with COVID-19 ARDS: a phase II randomized controlled trial. *Ann Intensive Care* 14:85. <https://doi.org/10.1186/s13613-024-01297-z>
 79. Morris IS, Bihari S, Goligher EC et al (2026) Monitoring respiratory effort during invasive mechanical ventilation: a multinational survey of clinical practice. *Intensive Care Med* 52:596–598. <https://doi.org/10.1007/s00134-025-08256-7>
 80. Lv Y, Dong M, Song H et al (2025) Respiratory effort monitoring: a novel, bedside, non-invasive, real-time method. *Crit Care* 29:272. <https://doi.org/10.1186/s13054-025-05514-4>
 81. Plens GM, Morais CCA, Gregol T et al (2026) Artificial intelligence algorithm to monitor inspiratory muscle effort and patient-ventilator dyssynchrony during mechanical ventilation. *Crit Care Med*. <https://doi.org/10.1097/CCM.0000000000007145>
 82. Cornejo RA, Brito R, Rivera C et al (2025) Influence of ventilatory settings on pendelluft and expiratory muscle activity in hypoxemic patients resuming spontaneous breathing. *Crit Care* 30:13. <https://doi.org/10.1186/s13054-025-05784-y>
 83. Burger-Klepp U, Maleczek M, Ristl R et al (2025) Using a clinical decision support system to reduce excess driving pressure: the ALARM trial. *BMC Med* 23:52. <https://doi.org/10.1186/s12916-025-03898-2>
 84. Weaver L, Shamohammadi H, Saffaran S et al (2024) Digital twins of acute hypoxemic respiratory failure patients suggest a mechanistic basis for success and failure of noninvasive ventilation. *Crit Care Med* 52:e473–e484. <https://doi.org/10.1097/CCM.0000000000006337>
 85. Goligher EC, Combes A, Brodie D et al (2019) Determinants of the effect of extracorporeal carbon dioxide removal in the SUPERNOVA trial: implications for trial design. *Intensive Care Med* 45:1219–1230. <https://doi.org/10.1007/s00134-019-05708-9>
 86. Ware LB, Matthay MA, Mebazaa A (2020) Designing an ARDS trial for 2020 and beyond: focus on enrichment strategies. *Intensive Care Med* 46:2153–2156. <https://doi.org/10.1007/s00134-020-06232-x>
 87. Bassi TG, Rohrs EC, Fernandez KC et al (2021) Brain injury after 50 h of lung-protective mechanical ventilation in a preclinical model. *Sci Rep* 11:5105. <https://doi.org/10.1038/s41598-021-84440-1>